

DIMENSIONALITY REDUCTION FOR K-MEANS AND LOW RANK APPROXIMATION

Michael Cohen, Sam Elder, Cameron Musco,
Christopher Musco, Mădălina Persu

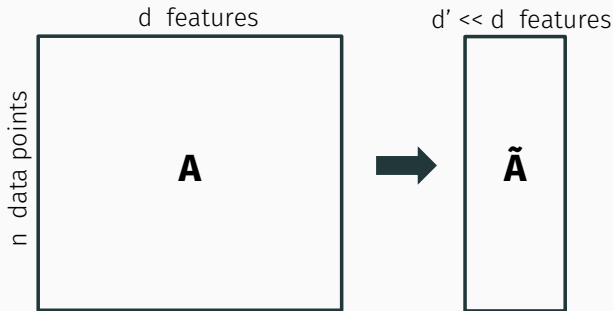
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Simple techniques to accelerate algorithms for:

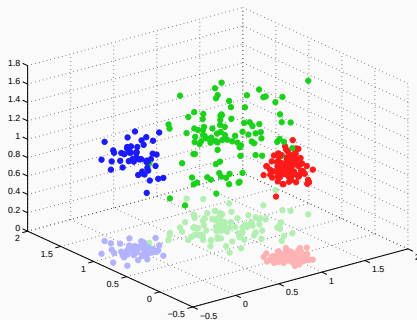
- k-means clustering
- principal component analysis (PCA)
- **constrained low rank approximation**

DIMENSIONALITY REDUCTION

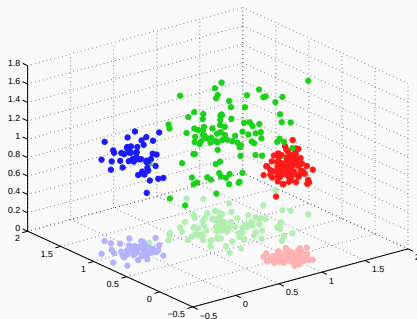
Replace large, high dimensional dataset with low dimensional sketch.



Solution on sketch $\tilde{\mathbf{A}}$ should approximate original solution.



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- Simultaneously improves runtime, memory requirements, communication cost, etc.

- Extremely common objective function for clustering



K-MEANS CLUSTERING

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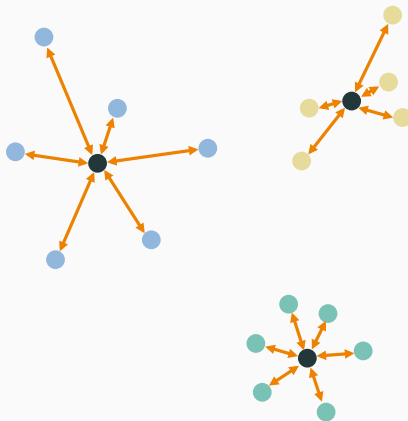
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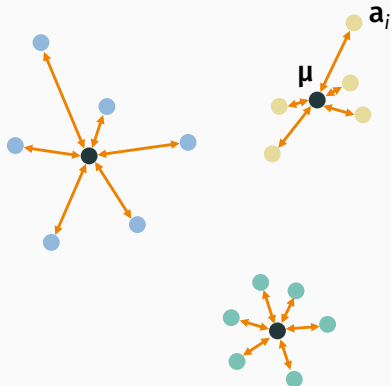


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K-MEANS CLUSTERING



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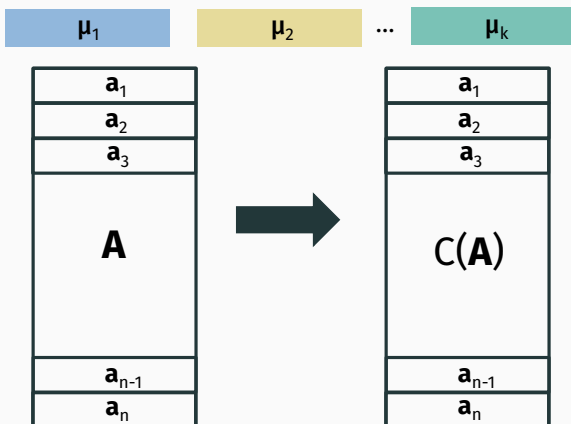
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Dimensionality reduction can speed up any of these algorithms.

k-means clustering == low rank approximation

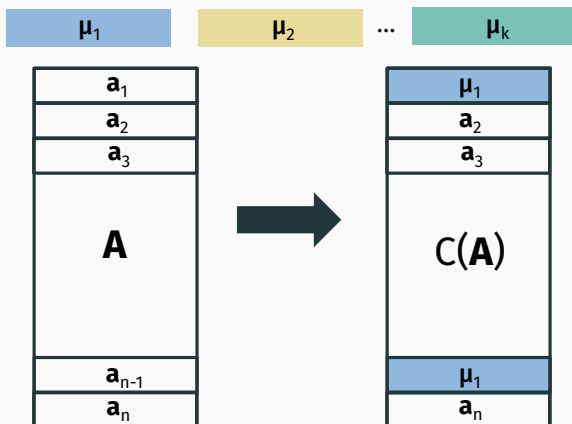
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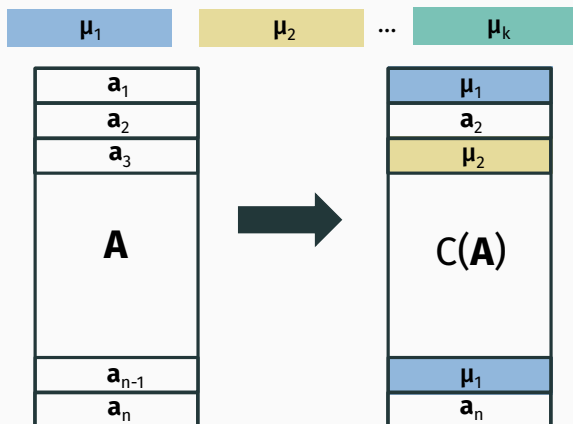
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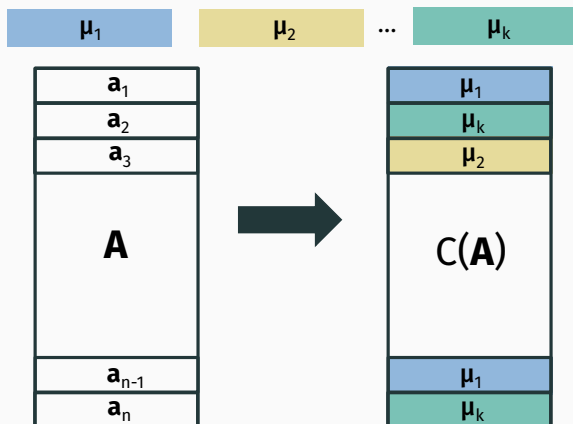
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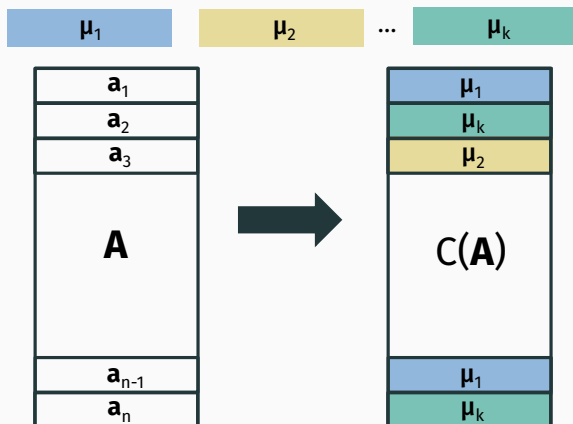
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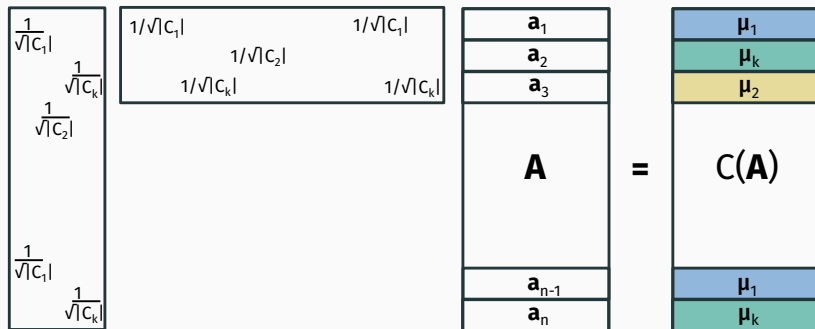
$$\min \sum_{i=1}^n \| \mathbf{a}_i - \boldsymbol{\mu}(\mathbf{a}_i) \|_2^2 = \| \mathbf{A} - \mathbf{C}(\mathbf{A}) \|_F^2$$



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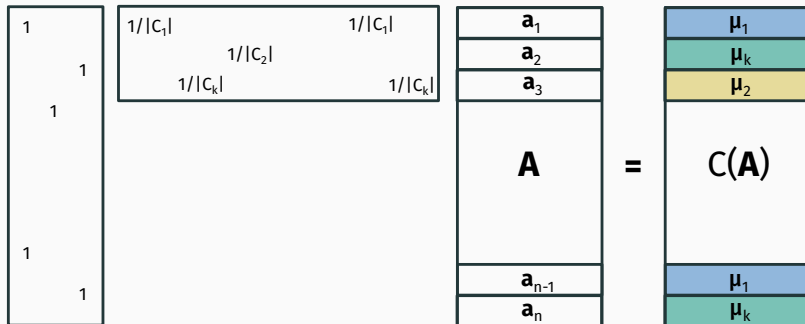
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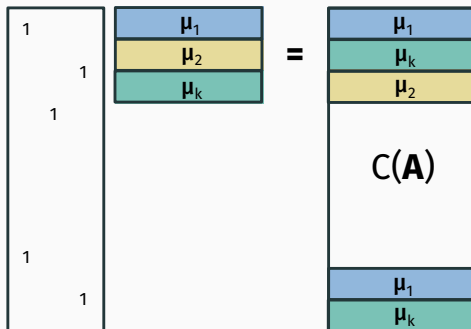
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$$\begin{array}{|c|} \hline \frac{1}{\sqrt{|C_1|}} \\ \hline \frac{1}{\sqrt{|C_k|}} \\ \hline \frac{1}{\sqrt{|C_2|}} \\ \hline \mathbf{X}_C \\ \hline \frac{1}{\sqrt{|C_1|}} \\ \hline \frac{1}{\sqrt{|C_k|}} \\ \hline \end{array}
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 \begin{array}{|c|} \hline \mu_1 \\ \hline \mu_k \\ \hline \mu_2 \\ \hline \mathbf{C}(\mathbf{A}) \\ \hline \mu_1 \\ \hline \mu_k \\ \hline \end{array}$$

$$\min_C \sum_{i=1}^n \|\mathbf{a}_i - \boldsymbol{\mu}(C[\mathbf{a}_i])\|_2^2 \implies \min_{\text{rank}(\mathbf{X})=k, \mathbf{X} \in \mathcal{S}} \|\mathbf{A} - \mathbf{X}\mathbf{X}^\top \mathbf{A}\|_F^2$$

Where $\mathbf{X}$ is a rank k orthonormal matrix and for k -means $\mathcal{S}$ is the set of all clustering indicator matrices.

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- Set $\mathcal{S} = \{\text{All rank } k \text{ orthonormal matrices}\}$ for principal component analysis/singular value decomposition.
- General form for **constrained low rank approximation**.

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$$\left\| \underbrace{\mathbf{A}}_d - \mathbf{X}\mathbf{X}^\top \mathbf{A} \right\|_F^2 \approx \left\| \underbrace{\tilde{\mathbf{A}}}_{O(k)} - \mathbf{X}\mathbf{X}^\top \tilde{\mathbf{A}} \right\|_F^2$$

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Projection-Cost Preserving Sketch

Specifically, we want:

$$\text{for all } \mathbf{X}, \|\tilde{\mathbf{A}} - \mathbf{X}\mathbf{X}^\top \tilde{\mathbf{A}}\|_F^2 \approx \|\mathbf{A} - \mathbf{X}\mathbf{X}^\top \mathbf{A}\|_F^2$$

Specifically, we want:

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If we find an $\mathbf{X}$ that gives:

$$\|\tilde{\mathbf{A}} - \mathbf{X}\mathbf{X}^\top \tilde{\mathbf{A}}\|_F^2 \leq \gamma \|\tilde{\mathbf{A}} - \widetilde{\mathbf{X}\mathbf{X}}_{opt}^\top \tilde{\mathbf{A}}\|_F^2$$

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then:

$$\|\mathbf{A} - \mathbf{X}\mathbf{X}^\top \mathbf{A}\|_F^2 \leq \gamma \cdot (1 + \epsilon) \|\mathbf{A} - (\mathbf{X}\mathbf{X})_{opt}^\top \mathbf{A}\|_F^2$$

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See **coresets** of Feldman, Schmidt, Sohler '13.

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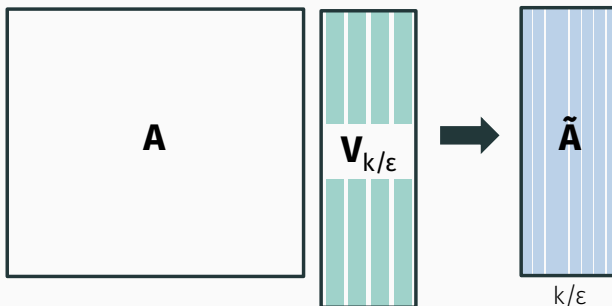
- k -means clustering is just constrained k rank approximation.
- We can construct a projection-cost preserving sketch $\tilde{\mathbf{A}}$ that approximates the distance from $\mathbf{A}$ to any rank k subspace.
- Stronger guarantee than has been sought in prior work on approximate PCA via sketching

What techniques give $(1 + \epsilon)$ projection-cost preservation?

	Previous Work			Our Results	
Technique	Reference	Dimensions	Error	Dimensions	Error
SVD	Feldman, Schmidt, Sohler '13	$O(k/\epsilon^2)$	$1 + \epsilon$	$\lceil k/\epsilon \rceil$	$1 + \epsilon$
Approximate SVD	Boutsidis, Drineas, Mahoney, Zouzias '11	k	$2 + \epsilon$	$\lceil k/\epsilon \rceil$	$1 + \epsilon$
Random Projection	"	$O(k/\epsilon^2)$	$2 + \epsilon$	$O(k/\epsilon^2)$ $O(\log k/\epsilon^2)$	$1 + \epsilon$ $9 + \epsilon$
Column Sampling	"	$O(k \log k/\epsilon^2)$	$3 + \epsilon$	$O(k \log k/\epsilon^2)$	$1 + \epsilon$
Deterministic Column Selection	Boutsidis, Magdon- Ismail '13	$r > k$	$O(n/r)$	$O(k/\epsilon^2)$	$1 + \epsilon$
Non-oblivious Projection	NA	NA	NA	$O(k/\epsilon)$	$1 + \epsilon$

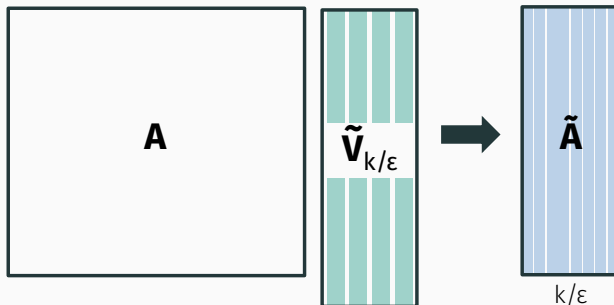
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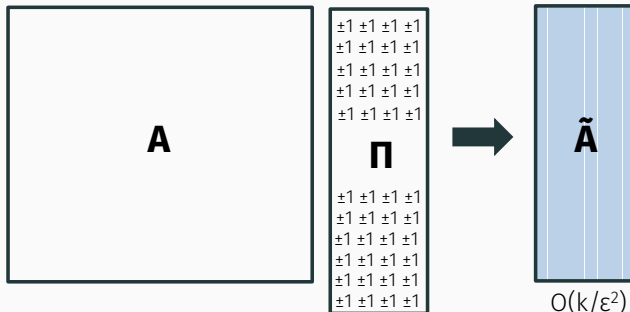


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- Practically very useful for k-means
- No constant factors on k/ϵ , and typically many fewer dimensions are required (Kappmeier, Schmidt, Schmidt '15)

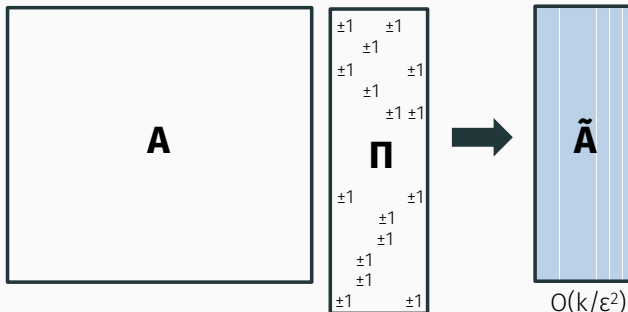
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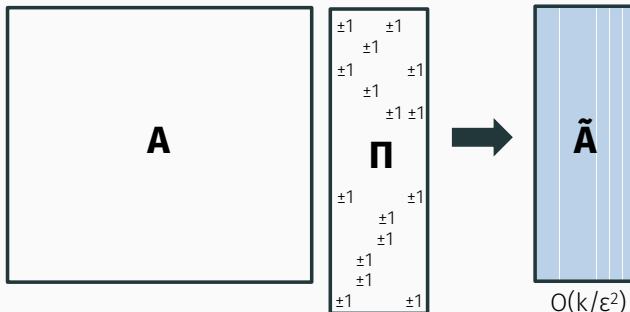
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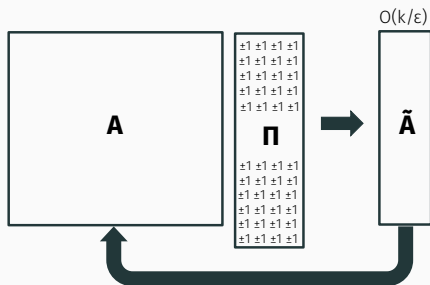
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- First sketch with dimension sublinear in k . Can $(9 + \epsilon)$ be improved?
- Sketch is data **oblivious**
 - Lowest communication distributed k -means (improves on Balcan, Kanchanapally, Liang, Woodruff '14)
 - Streaming principal component analysis in a single pass

Standard sketches for low rank approximation (Sarlós '06, Clarkson, Woodruff '13, etc):

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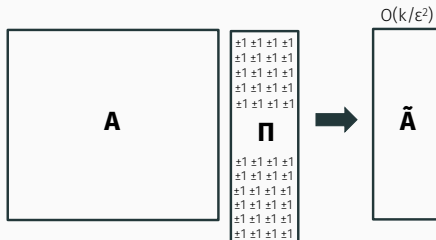


$$\|A - (P_{\tilde{A}}A)_k\|_F^2 \leq (1 + \epsilon)\|A - A_k\|_F^2.$$

$\text{span}(\tilde{A})$ contains a good low rank approximation for A , but we must return to A to find it.

Projection-cost preserving sketch for low rank approximation:

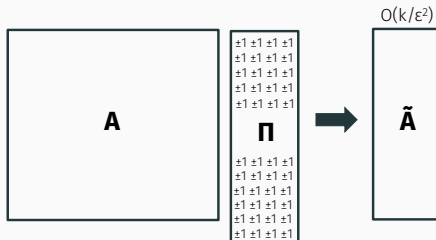
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Additional ϵ dependence for stronger sketch. **Is it required for oblivious approximate PCA?** (See Ghashami, Liberty, Phillips, Woodruff '14/15)

OUR RESULTS

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Non-oblivious Randomized Projection	Sarlós '06	NA	NA	$O(k/\epsilon)$	$1 + \epsilon$

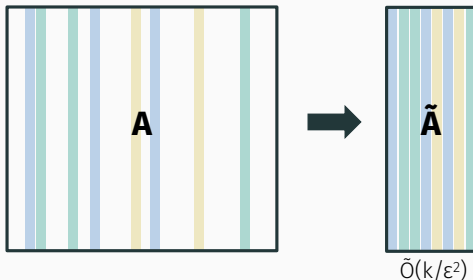


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- Very good practical performance: can get away with narrower Π than approximate SVD, although final point dimension is a constant factor higher.

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- $\mathbf{A}$ not only contains a small set of columns that span a good low rank approximation, but a small (reweighted) set whose top principal components approximate those of $\mathbf{A}$.
- First single shot sampling based dimensionality reduction for $(1 + \epsilon)$ error low rank approximation.

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- **A** not only contains a small set of columns that span a good low rank approximation, but a small (reweighted) set whose top principal components approximate those of **A**.
- First single shot sampling based dimensionality reduction for $(1 + \epsilon)$ error low rank approximation.
 - Applications to streaming column subset selection? Iterative sampling algorithms for the SVD?

Singular Value Decomposition (SVD):

The diagram illustrates the Singular Value Decomposition (SVD) of a matrix A . It shows the equation $A = U \Sigma V^T$ using visual representations of the matrices:

- Matrix A :** A white rectangle with the label A in the center.
- Equality:** An equals sign $=$ is placed between the matrices.
- Matrix U :** A rectangle with vertical green stripes, representing an orthogonal matrix.
- Matrix Σ :** A square with a yellow diagonal band. The diagonal elements are labeled $\sigma_1, \sigma_2, \dots, \sigma_{d-1}, \sigma_d$, representing the singular values.
- Matrix V^T :** A rectangle with horizontal blue stripes, representing an orthogonal matrix.

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- Matrix U :** A rectangle with vertical green stripes, representing orthonormal columns.
- Matrix Σ :** A rectangle with a yellow diagonal band containing the singular values $\sigma_1, \sigma_2, \dots, \sigma_{d-1}, \sigma_d$ in descending order.
- Matrix V^T :** A rectangle with horizontal blue stripes, representing orthonormal rows.

U, V have orthonormal columns. $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d$.

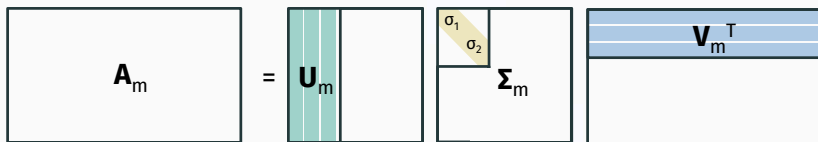
Singular Value Decomposition (SVD):

The diagram illustrates the Singular Value Decomposition (SVD) of a matrix A . It shows the equation $A = U \Sigma V^T$. Matrix A is represented by a light gray rectangle. Matrix U is represented by a rectangle with vertical green stripes. Matrix Σ is represented by a rectangle with a yellow diagonal band containing the singular values $\sigma_1, \sigma_2, \dots, \sigma_{d-1}, \sigma_d$. Matrix V^T is represented by a rectangle with horizontal blue stripes.

U, V have orthonormal columns. $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d$.

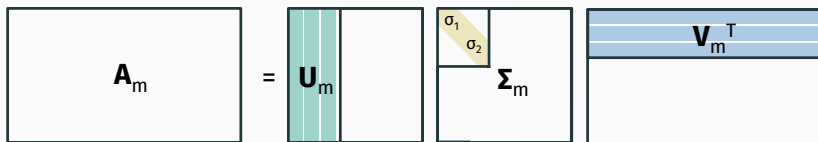
Same as Principal Component Analysis (PCA) if we mean center A 's columns/rows.

Partial Singular Value Decomposition (SVD):



$$\|A - A_m\|_F^2 = \min_{\text{rank}(X)=m} \|A - XX^T A\|_F^2$$

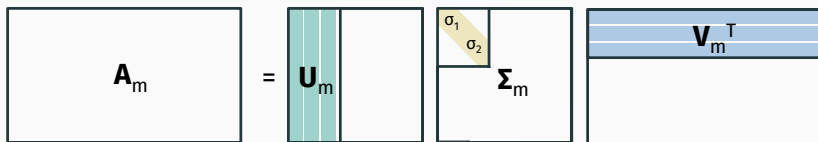
Partial Singular Value Decomposition (SVD):



$$\|A - A_m\|_F^2 = \min_{\text{rank}(X)=m} \|A - XX^T A\|_F^2$$

Claim: $\|A_{k/\epsilon} - XX^T A_{k/\epsilon}\|_F^2 + c = (1 \pm \epsilon) \|A - XX^T A\|_F^2$

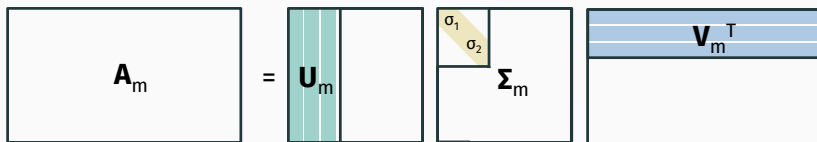
Partial Singular Value Decomposition (SVD):



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We work with A_m for simplicity. Denote $(A - A_m)$ as $A_{\setminus m}$

Claim: $\|\mathbf{A}_{k/\epsilon} - \mathbf{X}\mathbf{X}^\top \mathbf{A}_{k/\epsilon}\|_F^2 + c = (1 \pm \epsilon)\|\mathbf{A} - \mathbf{X}\mathbf{X}^\top \mathbf{A}\|_F^2$

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Split into (row) orthogonal pairs:

$$\boxed{\mathbf{A}} = \boxed{\mathbf{A}_{k/\epsilon}} + \boxed{\mathbf{A}_{\setminus k/\epsilon}}$$

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In words, the projection cost for $\mathbf{A}$ is explained by the cost over $\mathbf{A}_{k/\epsilon}$ plus the cost over $\mathbf{A}_{\setminus k/\epsilon}$. We want to argue that ignoring the tail term is fine.

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Need to show that:

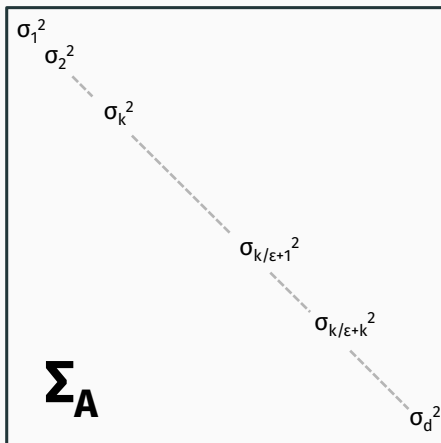
$$\|\mathbf{X}\mathbf{X}^\top \mathbf{A}_{\setminus k/\epsilon}\|_F^2 \leq \epsilon \|(\mathbf{I} - \mathbf{X}\mathbf{X}^\top) \mathbf{A}\|_F^2$$

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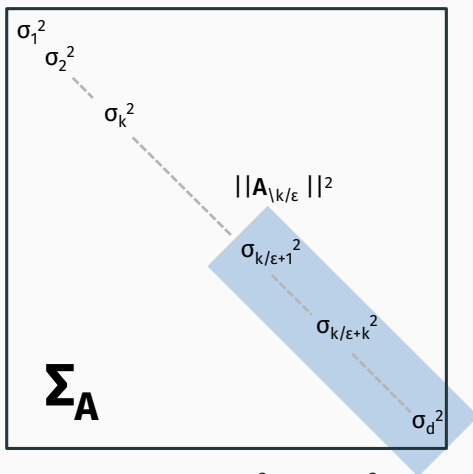
$$\|\mathbf{X}\mathbf{X}^\top \mathbf{A}_{\setminus k/\epsilon}\|_F^2 \leq \epsilon \|\mathbf{A} - \mathbf{A}_k\|_F^2$$

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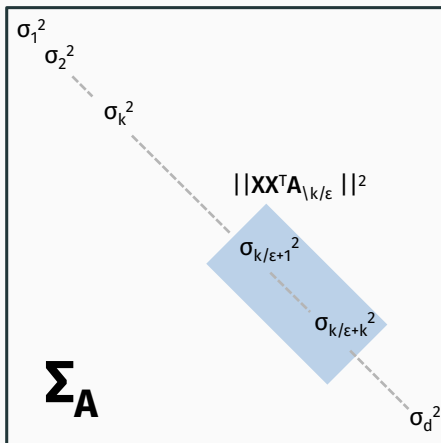
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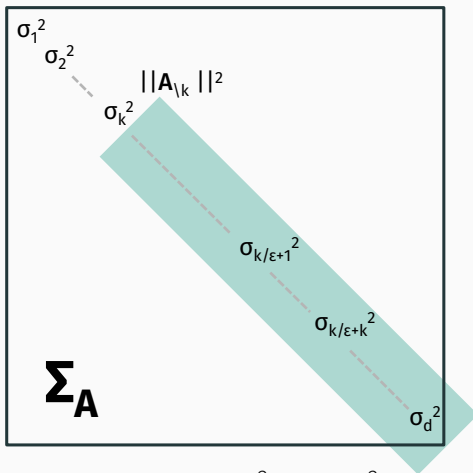
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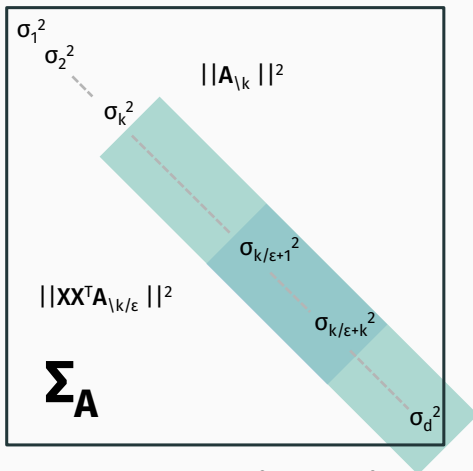
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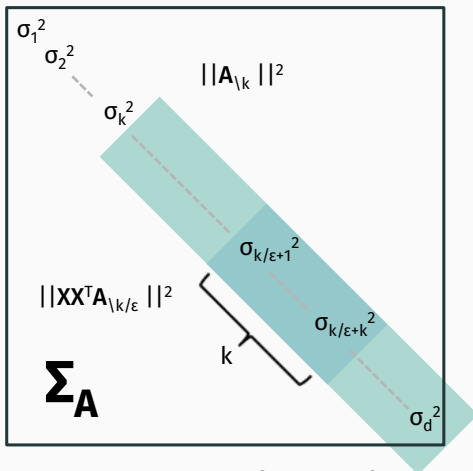
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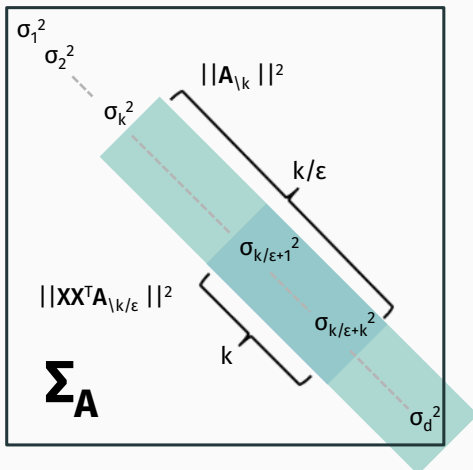
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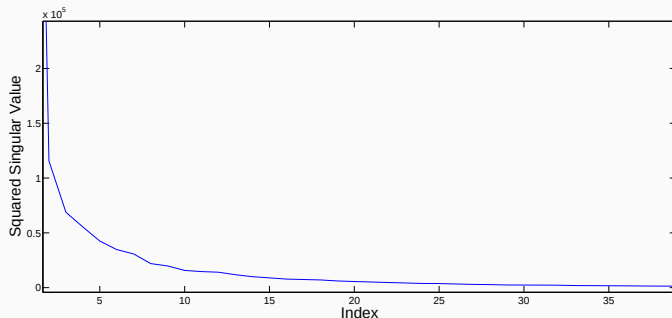


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All proofs have similar flavor:

$$\|\mathbf{A}_k - \mathbf{X}\mathbf{X}^\top \mathbf{A}_k\|_F^2 + \|\mathbf{A}_{\setminus k} - \mathbf{X}\mathbf{X}^\top \mathbf{A}_{\setminus k}\|_F^2$$

vs.

$$\|\mathbf{A}_k - \mathbf{X}\mathbf{X}^\top \mathbf{A}_k \mathbf{\Pi}\|_F^2 + \|\mathbf{A}_{\setminus k} - \mathbf{X}\mathbf{X}^\top \mathbf{A}_{\setminus k} \mathbf{\Pi}\|_F^2$$

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Except that:

- We have to worry about cross terms

$$\|\mathbf{A}\mathbf{\Pi} - \mathbf{X}\mathbf{X}^\top \mathbf{A}\mathbf{\Pi}\|_F^2 \neq \|\mathbf{A}_k \mathbf{\Pi} - \mathbf{X}\mathbf{X}^\top \mathbf{A}_k \mathbf{\Pi}\|_F^2 + \|\mathbf{A}_{\setminus k} \mathbf{\Pi} - \mathbf{X}\mathbf{X}^\top \mathbf{A}_{\setminus k} \mathbf{\Pi}\|_F^2$$

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- We have some hope of approximating $\|\mathbf{A}_{\setminus k} - \mathbf{X}\mathbf{X}^\top \mathbf{A}_{\setminus k}\|_F^2$

Rely on standard sketching tools:

- Approximate Matrix Multiplication: $\|\mathbf{A}\mathbf{\Pi}\mathbf{\Pi}^T\mathbf{B}\|_F^2 \leq \epsilon\|\mathbf{A}\|_F\|\mathbf{B}\|_F$
- Subspace Embedding: $\|\mathbf{Y}\mathbf{A}\mathbf{\Pi}\|_F^2 = (1 \pm \epsilon)\|\mathbf{Y}\mathbf{A}\|_F^2$ for all $\mathbf{Y}$ if $\mathbf{A}$ has rank k
- Frobenius Norm Preservation: $\|\mathbf{A}\mathbf{\Pi}\|_F^2 = (1 \pm \epsilon)\|\mathbf{A}\|_F^2$

See paper for the details!

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- Standard proof approach: Split $\mathbf{A}$ into orthogonal pairs. Top of spectrum can be preserved multiplicatively, only top singular values of bottom of spectrum matter.

PROOF SKETCH FOR SUBLINEAR RESULT

Split using *optimal* clustering:

The diagram illustrates the splitting of a matrix A into two components, $C^*(A)$ and $A - C^*(A)$, using optimal clustering. The matrix A is represented as a vertical stack of rows. The top three rows are labeled a_1 , a_2 , and a_3 . The bottom two rows are labeled a_{n-1} and a_n . The matrix $C^*(A)$ is represented as a vertical stack of rows. The top three rows are labeled μ_1 (blue), μ_k (teal), and μ_2 (yellow). The bottom two rows are labeled μ_1 (blue) and μ_k (teal). The matrix $A - C^*(A)$ is represented as a vertical stack of rows. The top three rows are labeled e_1 (blue), e_k (teal), and e_2 (yellow). The bottom two rows are labeled e_1 (blue) and e_k (teal). The equation is shown as $A = C^*(A) + A - C^*(A)$.

$$\begin{array}{|c|} \hline a_1 \\ \hline a_2 \\ \hline a_3 \\ \hline A \\ \hline a_{n-1} \\ \hline a_n \\ \hline \end{array} = \begin{array}{|c|} \hline \mu_1 \\ \hline \mu_k \\ \hline \mu_2 \\ \hline C^*(A) \\ \hline \mu_1 \\ \hline \mu_k \\ \hline \end{array} + \begin{array}{|c|} \hline e_1 \\ \hline e_k \\ \hline e_2 \\ \hline A - C^*(A) \\ \hline e_1 \\ \hline e_k \\ \hline \end{array}$$

PROOF SKETCH FOR SUBLINEAR RESULT

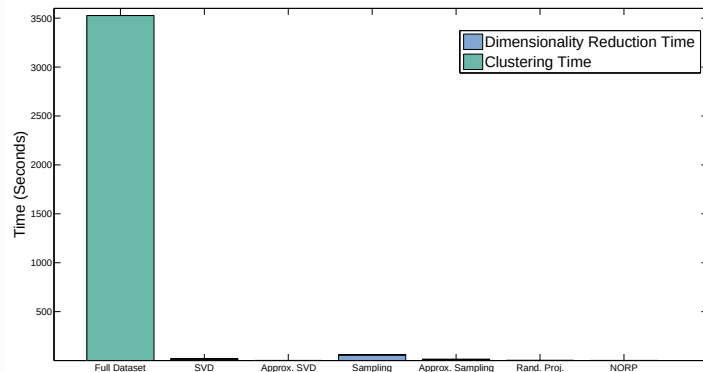
Split using *optimal* clustering:

The diagram shows the decomposition of a matrix A into two parts, $C^*(A)$ and $A - C^*(A)$, based on an optimal clustering of its rows. The matrix A is represented as a vertical stack of rows. The top three rows are labeled a_1, a_2, a_3 , and the bottom two are labeled a_{n-1}, a_n . The matrix $C^*(A)$ is shown as a vertical stack of rows, with the top three rows labeled μ_1, μ_k, μ_2 and the bottom two rows labeled μ_1, μ_k . The matrix $A - C^*(A)$ is shown as a vertical stack of rows, with the top three rows labeled e_1, e_k, e_2 and the bottom two rows labeled e_1, e_k . The equation $A = C^*(A) + A - C^*(A)$ is shown with the matrices arranged horizontally and separated by an equals sign and a plus sign.

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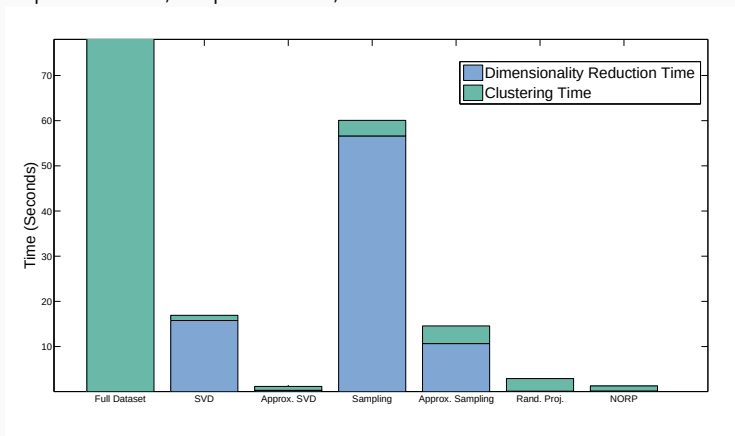
- $C^*(A)$ can be approximated with $O(\log k / \epsilon^2)$ dimensions.
- Not row orthogonal – have to use triangle inequality which leads to the $(9 + \epsilon)$ factor.

Experiments, implements, etc. in Cameron Musco '15



Time for $\epsilon = .01$ compared to baseline Lloyd's w/ k-means++.

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Thank you!